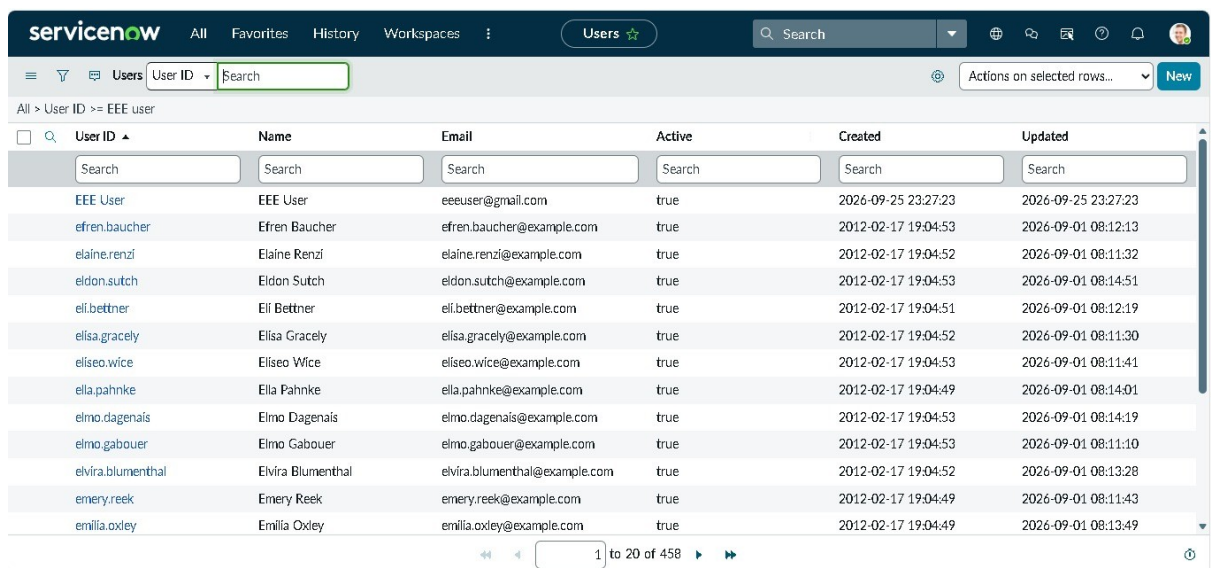


Script-Controlled ACL – Restrict Record Access Based on Field

Milestone-1 Creation of Users and Roles

Users Creation

- Log in to the ServiceNow instance with admin access.
- Create a test user:
- Navigate to User Administration → Users
- Click New
- User ID: EEE User
- First name: EEE
- Last name: User
- Email: eeeuser@gmail.com
- Save the record



User ID	Name	Email	Active	Created	Updated
EEE User	EEE User	eeeuser@gmail.com	true	2026-09-25 23:27:23	2026-09-25 23:27:23
efren.baucher	Efrén Baucher	efren.baucher@example.com	true	2012-02-17 19:04:53	2026-09-01 08:12:13
elaine.renzi	Elaine Renzi	elaine.renzi@example.com	true	2012-02-17 19:04:52	2026-09-01 08:11:32
eldon.sutch	Eldon Sutch	eldon.sutch@example.com	true	2012-02-17 19:04:53	2026-09-01 08:14:51
eli.bettner	Eli Bettner	eli.bettner@example.com	true	2012-02-17 19:04:51	2026-09-01 08:12:19
elisa.gracely	Elisa Gracely	elisa.gracely@example.com	true	2012-02-17 19:04:52	2026-09-01 08:11:30
eliseo.wice	Eliseo Wice	eliseo.wice@example.com	true	2012-02-17 19:04:53	2026-09-01 08:11:41
ella.pahnke	Ella Pahnke	ella.pahnke@example.com	true	2012-02-17 19:04:49	2026-09-01 08:14:01
elmo.dagenais	Elmo Dagenais	elmo.dagenais@example.com	true	2012-02-17 19:04:53	2026-09-01 08:14:19
elmo.gabouer	Elmo Gabouer	elmo.gabouer@example.com	true	2012-02-17 19:04:53	2026-09-01 08:11:10
elvira.blumenthal	Elvira Blumenthal	elvira.blumenthal@example.com	true	2012-02-17 19:04:52	2026-09-01 08:13:28
emery.reek	Emery Reek	emery.reek@example.com	true	2012-02-17 19:04:49	2026-09-01 08:11:43
emilia.oxley	Emilia Oxley	emilia.oxley@example.com	true	2012-02-17 19:04:49	2026-09-01 08:13:49

Roles Creation

- Navigate to User Administration → Roles
- Click New
- Name: Enter role name as bb1
- Description: Brief role purpose

- Click Submit
- Create other roles also like bb2,bb3,bb4.(follow the same procedure)
- Assign all the roles to the EEE User

servicenow All Favorites History Workspaces Roles Search

Roles Name Search

All > Name >= bb1

Name	Description	Elevated privilege
Search	Search	Search
bb1	Brief role purpose	false
bb2	Brief role purpose	false
bb3	Brief role purpose	false
bb4	Brief role purpose	false
bsm_legacy	Only users with the 'bsm_legacy' will see the module BSM Legacy within Dependencies View	false
bsm_legacy_admin	Only users with the 'bsm_legacy_admin' will see the module BSM Legacy within Dependencies View	false
business_process_admin		false
business_process_manager		false
business_process_user		false
business_relationship_manager	Person(s) responsible for both business strategy and business value in an effort to advance business objectives.	false
business_rule_admin	Can manage Business Rules	false
business_stakeholder	Use the Business Stakeholder role to provide authorized users to approve, view/read records across the organization and view reports. Note there is a fee associated with this role. Do not assign it to users without confirming your organization has the appropriate entitlement. Please review the documentation for additional details	false

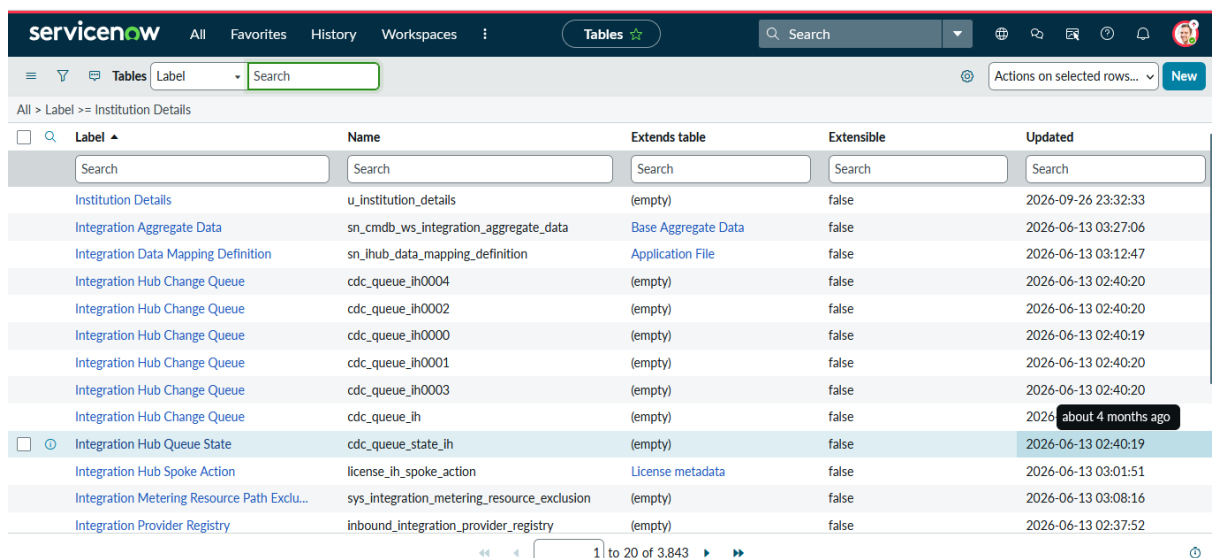
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Script-Controlled ACL-Restrict Record Access Based on Field Value

Milestone-2: Tables Creation

Creation of Tables

- In the Application Navigator, search for System Definition → Tables.
- Click New and create a custom table with the following details
- Label: Institution Details
- Name: u_institution_details
- Extends for : false`
- Save or Submit the form



The screenshot shows the ServiceNow interface for the 'Tables' list. The table has columns: Label, Name, Extends table, Extensible, and Updated. The 'Integration Hub Queue State' table is selected, showing its details.

Label	Name	Extends table	Extensible	Updated
Institution Details	u_institution_details	(empty)	false	2026-09-26 23:32:33
Integration Aggregate Data	sn_cmdb_ws_integration_aggregate_data	Base Aggregate Data	false	2026-06-13 03:27:06
Integration Data Mapping Definition	sn_ihub_data_mapping_definition	Application File	false	2026-06-13 03:12:47
Integration Hub Change Queue	cdc_queue_ih0004	(empty)	false	2026-06-13 02:40:20
Integration Hub Change Queue	cdc_queue_ih0002	(empty)	false	2026-06-13 02:40:20
Integration Hub Change Queue	cdc_queue_ih0000	(empty)	false	2026-06-13 02:40:19
Integration Hub Change Queue	cdc_queue_ih0001	(empty)	false	2026-06-13 02:40:20
Integration Hub Change Queue	cdc_queue_ih0003	(empty)	false	2026-06-13 02:40:20
Integration Hub Change Queue	cdc_queue_ih	(empty)	false	2026-06-13 02:40:19
Integration Hub Queue State	cdc_queue_state_ih	(empty)	false	2026-06-13 02:40:19
Integration Hub Spoke Action	license_ih_spoke_action	License metadata	false	2026-06-13 03:01:51
Integration Metering Resource Path Exclu...	sys_integration_metering_resource_exclusion	(empty)	false	2026-06-13 03:08:16
Integration Provider Registry	inbound_integration_provider_registry	(empty)	false	2026-06-13 02:37:52

- In the columns tab, user Insert a New row method to create the mentioned fields as shown below picture

- The following fields to the u_institution_details table
- Student Roll Number– Auto Number
- Student Name – Reference- User
- Faculty Name – Reference- User
- Branch – Choice (ECE, EEE, CSE)
- Email– String
- Phone Number– String
- Description– Multi String
- Save the Table

Table Institution Details

- Click on the Show List in the Related Links
- Create multiple records in the Student Records table with different branch values (ECE, EEE, CSE).

Script-Controlled ACL – Restrict Record Access Based on Field Value

Milestone-3: Creation of Access Control List- READ

ACL Configuration

- In the Application Navigator, search for System Security → Access Control (ACL).
- Elevate “Security_admin” role
- Click New to create a new ACL.
- Enter the following details
- Type: record
- Operation: read
- Name: u_institution_details
- Active: true
- Advanced: true
- In the Requires role related list, add: bb1 (custom role)
- Add Data condition as Branch is EEE
- In the Script field, add the following script
- Save the ACL form

Advanced Condition

Controlled by References 

Script ☐ Turn on ECMAScript 2021 (ES12) mode 



```
1 <function () {  
2     // Allow admin users full access  
3     if (gs.hasRole('admin')) {  
4         return true;  
5     }  
6     // Allow only EEE branch users to see EEE records  
7     if (gs.hasRole('bb1')){  
8         return true;  
9     }  
10    // Deny access for all others  
11    return false;  
12 }();
```

Requires role

   1 to 1 of 1   

	Role	
×	bb1	
+	Insert a new row...	

Script-Controlled ACL – Restrict Record Access Based on Field Value

Milestone-4: Creation of Access Control List-CREATE

Creation of ACL-Create

- Click New to create a new ACL(Create)
- Enter the following details
- Type: record
- Operation: create
- Name: u_institution_details
- Active: true
- In the Requires role related list, add: bb2 (custom role)
- No need to add any Data condition
- Save the ACL Record

The screenshot shows the ServiceNow interface for creating an Access Control List (ACL) record. The breadcrumb trail is 'Access Control - u_institution_details'. The form fields are as follows:

- * Type: record
- * Operation: create
- Decision Type: Allow If
- Admin overrides: ☒
- Protection policy: -- None --
- * Name: Institution Details [u_institution_details]
- Description: (empty text area)
- Applies To: No. of records matching the condition: 3
- Buttons: Add Filter Condition, Add OR Clause
- Fields: -- choose field --, -- oper --, -- value --
- Application: Global
- Active: ☒
- Advanced: ☐

Below the form is the 'Requires role' section, which contains a table with one role listed:

Role
bb2
insert a new row...

Script-Controlled ACL – Restrict Record Access Based on

Milestone 5-Creation of Access Control List-WRITE

Creation of ACL-Write

- Click New to create a new ACL(Write)
- Enter the following details
- Type: record
- Operation: Write
- Name: u_institution_details
- Active: true
- In the Requires role related list, add: bb3 (custom role)
- No need to add any Data condition
- Save the ACL Record

The screenshot shows the ServiceNow interface for creating an Access Control List (ACL) record. The page title is "Access Control - u_institution_details". The form includes the following fields and options:

- Type:** record
- Operation:** write
- Decision Type:** Allow If
- Application:** Global
- Active:** ☒
- Advanced:** ☐
- Admin overrides:** ☒
- Protection policy:** -- None --
- Name:** Institution Details [u_institution_details]
- Description:** (empty text area)
- Applies To:** No. of records matching the condition: 3
 - Add Filter Condition:** (button)
 - Add OR Clause:** (button)
 - Field:** -- choose field --
 - Operator:** -- oper --
 - Value:** -- value --

Requires role		   1 to 1 of 1   	
Role			
	bb3		
	insert a new row...		

Script-Controlled ACL – Restrict Record Access Based on Field value

Milestone-6:Creation of Access Control List-DELETE

Creation of ACL-DELETE

- Click New to create a new ACL(Delete)
- Enter the following details
- Type: record
- Operation: Delete
- Name: u_institution_details
- Active: true
- In the Requires role related list, add: bb4 (custom role)
- No need to add any Data condition
- Save the ACL Record

The screenshot shows the ServiceNow 'Access Controls' configuration page for a record named 'u_institution_details'. The interface includes a top navigation bar with 'servicenow' and various tabs like 'All', 'Favorites', 'History', and 'Workspaces'. The main configuration area contains several fields: 'Type' is set to 'record', 'Operation' is 'delete', 'Decision Type' is 'Allow If', 'Application' is 'Global', 'Active' is checked, and 'Admin overrides' is also checked. The 'Protection policy' is set to '-- None --'. The 'Name' field is 'Institution Details [u_institution_details]'. There is a 'Description' text area and an 'Applies To' section with buttons for 'Add Filter Condition' and 'Add OR Clause'. Below the configuration area is a 'Requires role' section with a table listing roles. The table has one row with the role 'bb4' and a red 'X' icon. A '+ insert a new row...' button is at the bottom of the table.

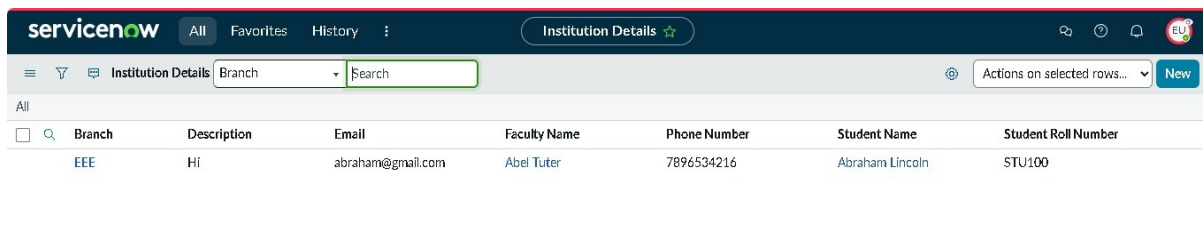
Role
X bb4
+ insert a new row...

Script-Controlled ACL – Restrict Record Access Based on Field Value

Milestone-7: Conclusion

Project Outcome

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces



Institution Details						
Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE	Hi	abraham@gmail.com	Abel Tutor	7896534216	Abraham Lincoln	STU100